

LETTER TO THE EDITOR

Response to Letter to the Editor by Eisenach and Yaksh on “Successful Pain Management with Epidural Oxytocin”

Miguel Condés-Lara,¹ Héctor Zayas-González,^{1,2} Alfredo Manzano-García¹ & Abimael González-Hernández¹

¹ Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Querétaro, México

² Hospital Regional de Petróleos Mexicanos, Guanajuato, México

Correspondence

A. González-Hernández, Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Boulevard Juriquilla 3001, Querétaro 76230, México.

Tel/Fax: +52-442-238-1042;

E-mail: abimaelgh@gmail.com or

abimaelgh@comunidad.unam.mx

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Response to Ethical concerns regarding human study,

We are grateful to Eisenach and Yaksh for their interest on our work [1]. Regarding their letter published in this issue [2], we have well noted their feedback and have some concerns with their comments.

First, we are in agreement in the awareness that in humans exposed to a novel or experimental treatment, researchers need to carefully attend the international and local ethic guidelines and rules of experimentation. For us, this is the case (we have the approval of our IRB, and we have the informed consent of the patients) and we are following those guidelines to continue with our scientific work. In addition, we (and all Editorials Boards) need to understand that in each case, the guidelines also should take into account all the circumstances in which a new treatment could improve a pathological state. Indeed, as stated by Elger and Hoppe [3], “the treatment does not follow a study protocol and if the patient provided informed consent, the physician is not only allowed, but obligated to apply approved (labeled and off-labeled) therapy.” In addition, we followed the four principles of Beauchamp and Childress, which indicate that patient–practitioner relationship should be guided by (1) autonomy, (2) beneficence, (3) nonmaleficence, and (4) justice [4].

Second, as you noticed in our report, although the patients had an adequate level of analgesia with the morphine plus ropivacaine regime, their quality of life was poor, and consequently, it is necessary to keep in mind that analgesia is not the unique factor when quality of life is evaluated. With the standard treatment (morphine plus ropivacaine), our patients presented a poor state of alertness and interaction with their relatives; in addition, the patients started to present serious side effects such as respiratory depression. In this context, it is important to point out that

oxytocin was administered to cancer patients in their terminal state, with a short life expectancy. Significantly, epidural administration of oxytocin plus ropivacaine gave our patients the same level of analgesia and additionally a better level of consciousness that allowed them to interact in a normal way with their relatives. We think that offering such outcome to a terminal patient is not only ethical but a priority.



Figure 1 Values of visual analogue scale (VAS) (A) and the Lattinen score (B) [6] obtained before and after (~20 min) epidural administration of oxytocin in seven cancer terminal ill patients with severe chronic pain. In all cases, the patients gave their informed consent, and the oxytocin-based treatment was followed after the poor analgesic response obtained with morphine (100 µg; i.v. continuous infusion). The oxytocin was given at dose of 2 IU (~4 µg) plus 0.1% ropivacaine in a volume of 4 mL. The values are expressed in mean ± standard deviation of seven patients.

Third, we are in accord with Eisenach and Yaksh that ropivacaine could play a role in the analgesia of the patients; nevertheless, the dose that we used was smaller than the reported in literature. In addition, it is necessary to give emphasis that the patients had the same level of analgesia in both regimes; it seems unlikely that in both cases the overall analgesia was given only by ropivacaine at subtherapeutic doses. Furthermore, although a mild neurolysis induced by the preservative in the oxytocin could be present, the overall effect observed could not be attributed to this effect. Certainly, the creation of formal guidelines on off-label use is necessary [5]. Indeed, the use of clinical experience with off-label drug use is an essential source for medical research [3] and our report provides strong evidence for the potential analgesic use of oxytocin in humans with severe pain.

Fourth, as noticed in the study, we are aware that larger and more robust-designed studies are necessary to establish unequivocally the analgesic properties of epidural oxytocin. Nevertheless, we think that this kind of short clinical reports is important for the scientific and medical community by providing with preliminary findings of the potential therapeutic use of different analgesic substances (carefully investigated in basic science protocols) and encourage research of such substances worldwide. In this context, we have seven cases

with similar positive results (Figure 1) and we think that we need to share this outcome with the scientific medical community.

Finally, our intention is to contribute to the scientific community with results intended to resolve medical challenges. We understand that our report will not provide a cure of the source of pain; instead, in this specific context, we are trying to provide a better life quality to suffering patients in the terminal states of life taking into account the principle *primum non nocere*. In any case, we appreciate the interest and encourage other biomedical scientific groups of research to continue, amplify, and re-evaluate the use of epidural oxytocin with larger cohorts of patients and other designs of studies. Nevertheless, we continue thinking that for us, it was relevant sharing our first clinical approach of the use of epidural oxytocin in pain to all of the medical community, and we encourage the editorial boards of scientific journals to publish these kinds of papers.

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